2.2.5.5 Mordant or Chrome Dyes

Natural dyes and some synthetic dyes do not have affinity towards fibres. With the help of some chemicals, they can be used to dye fibres. These chemicals are called mordants or mordant dyes. Mordant dyes have affinity for both fibre and dye and form a linkage between the dye molecule and the fibre.

2.2.5.6 Disperse Dyes

Disperse dyes are insoluble in water. Their solubility is increased by increasing temperature and by adding dispersing agents. They are suitable for dyeing hydrophobic fibres like nylon, polyester, acrylic and other synthetic fibres. Disperse dyes are non-ionic or neutral in nature. They have an excellent fastness to washing and sunlight exposure.

2.2.5.7 Vat Dyes

These dyes were initially fermented in a large wooden vessel known as vats and hence the name **"vat dyes".** They are insoluble in water but soluble in alkali. Vat dyes are accepted into the fibre in a reduced or vatted form. When it is re-oxidized, it is then fixed to the fibre firmly. The dyes are treated with a reducing agent that converts them into leuco compounds, which is soluble in alkalis. The process of making the dyes soluble is known as Vatting. The leuco compounds are either colourless or exhibit a colour that is different from the colour of final product, which is achieved after oxidation. Once the dye is attached to the fibre, it gets oxidised and changed into an insoluble colour product that gets trapped inside the fibre. Indigo, an example of vat dyes has been used in India for long periods.

In the soluble form, vat dyes have excellent affinity for cellulosic fibres. Vat dyes are the fastest dyes for cotton, linen and rayon. They may be also applied to wool, nylon, polyester and acrylics with the use of a mordant. Vat dyes are resistant to light, acids and alkalies.

The basic principles of vat dye application can be summarized as follows:

- Conversion of the insoluble vat dye into the soluble leuco form by reduction or "vatting".
- Absorption of the soluble reduced dye by the fibres
- Conversion of the absorbed dye back to the insoluble state by oxidation
- After treatment of the dye or printed material in a hot detergent bath to produce the true and stable shade with maximum fastness

2.2.5.8 Sulphur Dyes

Presence of sulphur is the characteristic feature of sulphur dyes. Like vat dyes, sulphur dyes are water insoluble and made soluble by the addition of reducing agents and alkali as solubiliser. They are converted into soluble leuco compounds by using sodium sulphide or sodium hydrosulphite as reducing agent. The leuco compounds have high affinity for cellulosic fibres. After dyeing the fabric, the dye is converted into insoluble form by the addition of potassium dichromate and acetic acid. Sulphur dyes do not produce brighter shades. They are cheap and produce colours like navy blue, black, khaki and olive green. They are used for black more than any other colours. They have excellent wash fastness property. The fabric loses its softness after dyeing with sulphur dyes.

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